STRUCON CONSULTING · DIGITAL INFRASTRUCTURE REVIEW

Website Rebuild: Comparative Audit & Business Case

An evidence-based comparison of the current **strucon.net** against the newly built platform — covering trust, typography, performance, SEO, lead generation, and security — with the published research that connects each detail to revenue.

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STR-WEB-2026-01	1 Aug 2026	A	For approval

METHOD

Live HTTP audit + build analysis

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01 · EXECUTIVE SUMMARY

The current site actively works against the business. The new one is built to win it.

We audited the live **strucon.net** over HTTP and compared it against the production build of the new platform. The findings below are **measured facts, not opinions** — and each is paired with published research quantifying what it costs in bounced visitors, lost rankings, and unearned distrust.

CMS CORE

Dec 2018

Current site runs WordPress 5.0.22 — a seven-year-old core with legacy WooCommerce & Slider Revolution plugins.

META DESCRIPTIONS FOUND

0

Not one page on the current site tells Google what it's about. The new site: unique metadata on every route.

STRUCTURED DATA (SCHEMA)

0 vs 6

Current site: zero JSON-LD objects. New site: Organization, LocalBusiness, Service, FAQ, Article, Breadcrumb.

HOMEPAGE SLIDER PAYLOAD

1.33 MB

Three unoptimized JPEGs (723 KB + 494 KB + 112 KB) vs AVIF/WebP with priority-loaded LCP on the new site.

SITEMAP.XML

404

The current site has no sitemap for Google to crawl. The new one generates it automatically from content.

LEAD-CAPTURE PATHS

1 vs 5

One generic contact page vs five intent-specific conversion flows with validation and tender upload.

THE ONE-SENTENCE CASE

Research says visitors judge a firm's credibility in **50 milliseconds**,[1] that **75% of users judge company credibility from website design**,[2] and that B2B buyers now complete roughly **two-thirds of their evaluation digitally before ever talking to a supplier**.[12] For most EPC prospects, strucon.net *is* the first impression and the shortlist decision — and today it presents 20+ years of engineering excellence through a 2018-era template with broken stat counters.

02 · METHODOLOGY & HONESTY NOTES

How this report was produced

- **Current site:** live audit of `strucon.net` — HTTP headers, full homepage HTML, font/script/stylesheet inventory, asset weights, robots/sitemap probes, and inner-page checks.
- **New site:** measured from the production build (44 routes, static/SSG output) and its verified behaviors — admin save-to-live cycle, form validation, schema output.
- **Research figures** are from published industry studies (cited inline, listed in Sources). They quantify *directional* effects observed across many sites — strong evidence, not guarantees for any single site.

- **Performance:** the new site is engineered for 95+ Lighthouse (static HTML, self-hosted fonts, AVIF/WebP, minimal JS). Final scores should be validated on production hosting — this report does not claim a measured score before that.

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03 · FIRST IMPRESSIONS & TRUST

Design is not decoration — it is the credibility audit every visitor runs

Stanford's Web Credibility Project — still the largest study of its kind (2,684 participants) — found that when people assess whether a company can be trusted, **the look and design of the site is the single most-cited factor, ahead of the actual information on it**.[2] A companion study on advice sites found **94% of user comments explaining mistrust were about design** — layout, typography, clutter — not content.[3]

50 ms

Time in which users form a stable aesthetic judgment of a webpage — before reading a single word.

Lindgaard et al., Behaviour & IT, 2006 [1]

75%

Of users admit judging a company's credibility by its website design.

Stanford Web Credibility Project [2]

94%

Of first-impression mistrust feedback relates to design factors rather than content.

Sillence et al., 2004 [3]

88%

Of online users are less likely to return to a site after a bad experience.

Gomez / Akamai research [4]

What the 50-millisecond judgment sees today

✕ CURRENT STRUCON.NET

- Generic construction template (`buildpro` theme, WPBakery page builder) — indistinguishable from thousands of small-contractor sites.
- Animated stat counters that render as **"0" Hours of Work, "0" Clients, "0" Staff** — the homepage literally displays zeros next to trust claims.
- **ISO 9001:2008** on the masthead — a standard withdrawn in September 2018. To a procurement auditor this reads as a lapsed certification.
- Footer: "Copyright 2023" — signals an unmaintained property.
- Stock slider carousel with 2019-era imagery.

✓ NEW PLATFORM

- A proprietary "engineering drawing" identity — blueprint grids, discipline codes (STR/DTL/BIM), title-block data strips. It could belong to no one but a structural engineering firm.
- Working stats bar, hero title-block with founding year, tonnage, and certification.
- Case studies in the format EPC buyers actually evaluate: **Challenge** → **Solution** → **Results** with tonnage, schedule, and code metrics.
- Every trust claim placeholder is flagged for verification before go-live — nothing false ships.
- Consistent CTAs on all 25+ pages, so credibility converts instead of evaporating.

Why this matters for EPC specifically: Gartner finds B2B buying groups spend only **17% of their journey meeting potential suppliers** — the rest is independent research, mostly online.[12] McKinsey reports **~70–80% of B2B decision-makers now prefer remote or digital self-serve interactions**.[13] When a Middle-East EPC engineer shortlists steel detailers at 11pm, the website *is* the meeting. The current site loses that meeting in 50 milliseconds.

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04 · TYPOGRAPHY — THE DEEP DIVE

Font-by-font: what each site says before you read it

Typography research is unambiguous: typefaces measurably change how *credible*, *effortless*, and *professional* content feels. In the New York Times "Baskerville experiment" (~45,000 participants), the same statement earned **measurably higher agreement when set in one typeface versus another** — readers found identical words more trustworthy purely from the letterforms.[5] Song & Schwarz showed that hard-to-process type makes described tasks feel harder and people less willing to act[6] — directly relevant when the "task" is engaging an engineering vendor. And an MIT

AgeLab / Monotype study found humanist letterforms were processed **faster than square-grotesque faces**, especially at a glance.[7]

TYPOGRAPHY SYSTEM — MEASURED COMPARISON

ATTRIBUTE	CURRENT STRUCON.NET	NEW PLATFORM
Typefaces	Montserrat (headings) + Open Sans (body) The default pairing of the purchased "buildpro" theme — among the most-used combinations on the web, so the site reads as a template even to non-designers.	Space Grotesk (display) + Inter (body) + IBM Plex Mono (data/annotations) A three-role system chosen for the subject: a technical display voice, a screen-optimized body face, and a monospace "drawing annotation" layer unique to an engineering firm.
Weights loaded	7 weights of Montserrat (200–800) + Open Sans 400, plus a *second duplicate* Google Fonts request Measured in the homepage HTML: two separate `fonts.googleapis.com` stylesheets. Most weights are never used — pure dead payload.	Only the weights used, subset to Latin Loaded via `next/font`, which subsets and serves exactly what the design consumes.
Delivery	`External CDN, render-blocking` Two round-trips to Google's servers before text can settle; classic flash-of-invisible/unstyled text on slow site-office connections.	`Self-hosted, zero external requests` `next/font` inlines and self-hosts with automatic fallback metrics (`size-adjust`) — text renders instantly and never shifts.
Layout shift from fonts	`Yes — swap reflow` Web-font swap without metric-matched fallbacks causes visible jumps; layout shift is a scored Core Web Vitals failure (CLS).	`None by construction` Metric-compatible fallbacks are generated at build time, so CLS from fonts is ~0.
Body legibility	Open Sans 400 at theme defaults Adequate but characterless; line lengths uncontrolled by the page builder.	Inter — tall x-height, open apertures, designed for screens Body text capped at ~68 characters per line, the readability optimum; fluid type scale via `clamp()` from mobile to 4K.
Distinctiveness	`Template vernacular` Montserrat/Open Sans signals "bought theme" — the opposite of a firm selling precision.	`Ownable identity` The IBM Plex Mono annotation layer (discipline codes, tonnage labels, title blocks) mirrors the fabrication drawings Strucon actually produces — typography as proof of domain.

+1.5%

Higher agreement with identical statements when set in a more credible typeface — from ~45,000 NYT readers. Small per reader; compounding across every visitor.

Errol Morris NYT experiment, 2012 [5]

2×

Tasks described in a hard-to-read font were estimated to take roughly twice as long — processing fluency shapes willingness to engage.

Song & Schwarz, Psychological Science, 2008 [6]

−10.6%

Reduction in glance time for humanist vs square-grotesque type in MIT AgeLab trials — legibility differences are measurable, not taste.

Reimer, Mehler & Monotype / MIT AgeLab, 2012 [7]

The business translation: a procurement director skimming ten vendor sites doesn't consciously note x-heights. They feel that one site was *effortless and precise* and the others were *noisy* — and fluency research shows that feeling transfers onto judgments of the company itself.[6] The new type system is engineered to make Strucon the effortless one.

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05 · PERFORMANCE & CORE WEB VITALS

Speed is a revenue line, and Google made it a ranking factor

MEASURED TECHNICAL PAYLOAD — HOMEPAGE

METRIC (MEASURED)	CURRENT STRUCON.NET	NEW PLATFORM
Rendering model	WordPress 5.0.22 + WooCommerce 3.6.7 + WPBakery + Slider Revolution, assembled per request	Next.js 16 — pages pre-rendered to static HTML at build time; nothing assembled on request
Script tags	`39`	`Minimal` — server components; JS only for nav, reveal animations, and forms
Stylesheets	`21`	`1` — Tailwind, purged of unused rules
Hero imagery	`1.33 MB` of JPEG sliders (723 + 494 + 112 KB), no modern formats	`AVIF/WebP` via `next/image`, responsive sizes, lazy-loaded below the fold, LCP asset priority-loaded
Fonts	2 external render-blocking CDN requests, 8 weights	Self-hosted, subset, zero external requests, zero CLS
E-commerce code	`WooCommerce loaded` on an engineering consultancy site with nothing for sale — pure overhead	None
Carousel	Slider Revolution 5.4.8.1 — heavy jQuery slider	No carousel (see §07 — carousels measurably underperform)

+32%

Increase in bounce probability as mobile load time grows from 1s to 3s; +90% by 5s.

Google/SOASTA machine-learning study, 2017 [8]

53%

Of mobile visits are abandoned if a page takes longer than 3 seconds.

Google mobile research, 2016 [9]

+8.4%

Conversion lift observed from a 0.1-second speed improvement (retail; +10.1% travel).

Deloitte × Google, "Milliseconds Make Millions", 2020 [10]

Ranking factor

Core Web Vitals (loading, interactivity, layout stability) are part of Google's ranking systems since the 2021 Page Experience update.

Google Search Central [11]

The current homepage asks a first-time visitor — often on a site-office connection or a phone — to download a seven-year-old plugin stack, 21 stylesheets, 39 scripts, and

1.33 MB of slider JPEGs before the page settles. The new site ships pre-rendered HTML with the heavy work done at build time. Same fibre, different physics.

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06 · SEARCH VISIBILITY

SEO head-to-head: the current site is nearly invisible to Google by choice

ON-SITE SEO INVENTORY — MEASURED

SIGNAL	CURRENT STRUCON.NET	NEW PLATFORM
Title tags	`One generic title` "STRUCON Consulting – CIVIL AND STRUCTURAL ENGINEERING" — no service, industry, or intent keywords.	`Unique per route` e.g. "Steel Detailing Company — Tekla Fabrication Drawings", "Power Plant Structural Design & Steel Detailing".
Meta descriptions	`0 found` Homepage and inner pages checked — none. Google invents snippets; CTR is left to chance.	`Every route` Commercial-intent copy written per page.
Structured data	`0 JSON-LD objects`	`6 schema types` Organization, LocalBusiness/ProfessionalService, Service, FAQPage, Article, BreadcrumbList — FAQ & Article are rich-result eligible.
sitemap.xml	`404 — does not exist`	`Auto-generated` Every new project, service, or blog post added via the admin panel appears automatically.
Open Graph / social cards	`0 OG tags` Links shared on LinkedIn — Strucon's primary channel — render bare.	`Full OG + Twitter cards` With a branded 1200×630 share image generated at build.
Keyword architecture	None discernible	Commercial-intent clusters mapped page-by-page "Steel Detailing Company", "Tekla Modeling Services", "Pipe Rack Design", "Conveyor Gallery Design", "EPC Engineering Design"…
Content engine	`No blog`	`Blog with Article schema` Seed posts targeting long-tail EPC queries; new posts publishable from the admin panel.
Internal linking	Flat menu only	Services ↔ industries ↔ projects ↔ blog cross-linked with breadcrumbs sitewide
Indexable pages	~15 thin pages	44 routes, each targeting a distinct query cluster

+30.8%

Average relative CTR gain from moving up a single organic position — rankings compound.

Backlinko CTR study, 4M results [14]

82%

Higher click-through rate observed on pages appearing as rich results vs plain results (Nestlé case study). Rich results require the structured data the current site has none of.

Google Search Central case studies [15]

44 vs ~15

Indexable, individually optimized pages — each new page is another way for a buyer with a tender in hand to find Strucon.

Measured build output

The strategic point: Strucon doesn't need to win "engineering" — it needs to win "*steel detailing company*", "*Tekla modeling services*", "*pipe rack design*": the queries typed by someone with a tender in hand. The current site doesn't target a single one of them anywhere in its markup. The new site dedicates an optimized page to each.

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07 · LEAD GENERATION

From brochure to acquisition machine

✕ CURRENT: ONE EXIT DOOR

- A single "Contact Us" page; no CTAs woven through content.
- Hero is a rotating carousel — Notre Dame's benchmark study found only **~1% of visitors ever click a carousel**, and 84% of those clicks are on slide one.[16]
- No way to submit a tender, book a review, or request a scoped proposal.
- No case-study format a buyer can forward to a colleague.

✓ NEW: FIVE INTENT-MATCHED PATHS

- **Request Consultation · Request Proposal · Upload Tender Documents (file upload) · Book Engineering Review · Download Company Profile** — each with its own validated form, fields matched to intent.
- Every one of the 25+ pages ends in a relevant CTA; the navbar carries a persistent consultation button.
- Case studies structured as Challenge → Solution → Results with metrics — the internal-forwarding format.
- Client-side + server-side validation (React Hook Form + Zod), graceful success states, and a lead API ready to wire to email/CRM.

$12M

Annual revenue gain Expedia attributed to deleting a single confusing form field — form friction has measurable cost, and every new-site form asks only what its intent requires.

Expedia / UX case study [17]

1.07%

Of visitors interact with homepage carousels at all — the current site spends its most valuable pixels on a pattern users ignore.

Runyon, University of Notre Dame [16]

17%

Of B2B buying-journey time is spent meeting suppliers; the website carries the rest of the evaluation.

Gartner B2B buying research [12]

ILLUSTRATIVE FUNNEL MATH — NOT A FORECAST

Suppose search + referrals bring **1,000 relevant visitors/month**. Industry-typical B2B service conversion runs ~1–2% on a well-built site and materially lower on a slow, CTA-less one. If the rebuild moves conversion from 0.5% to 1.5% — well within the range the cited speed, trust, and form research supports — that is **5 → 15 qualified enquiries/month**. If even one incremental enquiry per quarter becomes a detailing contract, the rebuild pays for itself many times over. The multiplier compounds again as the SEO architecture grows traffic itself.

08 · SECURITY & MAINTAINABILITY

The current stack is a documented breach risk

PLATFORM RISK PROFILE

FACTOR	CURRENT STRUCON.NET	NEW PLATFO
Core software	`WordPress 5.0.22` The 5.0 line shipped December 2018. Running years-old branches means missing hardening the ecosystem assumes.	`Next.js 16` Upgraded duri specifically to
Plugin surface	`WooCommerce 3.6.7 (2019) + WPBakery + Slider Revolution 5.4.8.1` Slider Revolution is historically among the most-exploited WordPress plugins — the 2014–15 RevSlider vulnerability compromised 100k+ sites (SoakSoak). Sucuri's incident reports consistently identify outdated plugins/themes as the leading WordPress infection vector. [18]	`No plugin e` A handful of au dependencies, in one place.
Attack surface	Dynamic PHP + database + admin exposed at a well-known URL	Static pre-rende admin is cookie- configured cred search indexing
Content editing	Full WP admin — powerful, dated, and risky to update (theme lock-in via page builder)	Purpose-built / Projects, Blog, Te Careers; change code and no plu
Cost of a hack	Beyond cleanup: Google flags compromised sites in results and browsers ("This site may be hacked"), destroy rankings this report is about. Risk is asymmetric — prevention is cheap, recovery is not. [18]	

09 · ACCESSIBILITY & REACH

Small details, larger market

- **WCAG-conscious build:** semantic landmarks, skip-to-content link, visible keyboard focus, form labels and error messaging, contrast-checked palette, `prefers-reduced-motion` respected. The current site's page-builder markup offers none of this systematically.
- **Why it's business, not charity:** the WHO estimates ~16% of the world lives with some disability; accessibility overlaps heavily with usability for everyone — including engineers on gloves-and-glare site tablets. Accessible structure is also what search crawlers parse best, so the SEO and accessibility work compound.
- **Global audience:** fluid type and mobile-first layout serve the low-bandwidth, small-screen contexts common across EPC site offices — precisely where the 1.33 MB slider payload fails today.

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10 · RECOMMENDATION

Replace, redirect, verify — in that order

Recommendation: replace strucon.net with the new platform. Every measured dimension — trust signals, typography, payload, search architecture, lead capture, security — favors the rebuild, and the research above prices each gap in lost enquiries.

Go-live conditions (the honest checklist)

- **301 redirect map** from every old URL (`/index.php/about/` etc.) to its new equivalent — preserves the domain's existing link equity.
- **Verify before publishing:** the stats bar (26+ years / 500+ projects / 100+ engineers / 20 countries), current ISO certification version, real client names/logos with display rights, real leadership bios, and NDA-cleared case-study details. Placeholders are flagged in the build and must not ship as fact.
- **Wire the lead API** to email/CRM and file storage; set a strong `ADMIN_PASSWORD` .
- **Validate Lighthouse/CWV on production hosting** and submit the new sitemap in Google Search Console.
- **Keep the hero video ≤ ~3 MB** (compressed, muted, with poster) so it never undoes the performance case.

SCOPE CAVEAT

Research statistics cited here describe effects measured across large samples of other sites; they establish direction and order of magnitude, not a guaranteed outcome for strucon.net. The measured audit facts (versions, payloads, missing tags) are directly verifiable today and are sufficient grounds for the recommendation on their own.

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11 · SOURCES

Research cited

1. **Lindgaard, Fernandes, Dudek & Brown (2006)** — "Attention web designers: You have 50 milliseconds to make a good first impression!", *Behaviour & Information Technology* 25(2).
2. **Stanford Web Credibility Project — Fogg et al. (2002)** — "How Do People Evaluate a Web Site's Credibility?", Stanford Persuasive Technology Lab; 2,684 participants. Widely reported as "75% of users judge credibility by design"; the study's coded data show visual design cues as the most frequently mentioned credibility factor (46.1% of comments).
3. **Sillence, Briggs, Fishwick & Harris (2004)** — "Trust and mistrust of online health sites", CHI 2004; 94% of mistrust comments related to design elements.
4. **Gomez / Akamai** — "Why Web Performance Matters" consumer research; 88% of online consumers less likely to return after a bad experience.
5. **Errol Morris / NYT (2012)** — "Hear, All Ye People; Hearken, O Earth" typeface-credibility experiment, ~45,000 respondents; Baskerville produced a statistically significant lift in agreement vs other faces.
6. **Song & Schwarz (2008)** — "If It's Hard to Read, It's Hard to Do", *Psychological Science* 19(10); processing fluency alters effort estimates and willingness.
7. **Reimer & Mehler, MIT AgeLab × Monotype (2012)** — glance-legibility trials; humanist typeface reduced response/glance time vs square grotesque (~10.6% in male drivers).
8. **Google / SOASTA (2017)** — mobile page-speed machine-learning study; bounce probability +32% (1→3s), +90% (1→5s), +123% (1→10s).
9. **Google (2016)** — mobile speed research; 53% of mobile site visits abandoned beyond 3 seconds.
10. **Deloitte × Google (2020)** — "Milliseconds Make Millions"; +0.1s speed → +8.4% retail / +10.1% travel conversions.
11. **Google Search Central** — Page Experience update (2021): Core Web Vitals in ranking systems; HTTPS, mobile-first indexing documentation.

12. **Gartner** — "The B2B Buying Journey": buying groups spend ~17% of journey time meeting suppliers; ~27% researching independently online.
13. **McKinsey (2020–21)** — B2B Pulse: 70–80% of B2B decision-makers prefer remote human interaction or digital self-service.
14. **Backlinko (Dean)** — organic CTR analysis of 4M+ Google results; +30.8% relative CTR moving up one position; #1 averages ~27–28%.
15. **Google Search Central case studies** — structured data outcomes: Nestlé (~82% higher CTR on rich-result pages), Rotten Tomatoes (+25% engagement), and others; rich-result eligibility requires valid structured data.
16. **Runyon, E., University of Notre Dame** — "Carousel Interaction Stats": ~1.07% of visitors clicked carousels; 84%+ of clicks on the first slide. Corroborated by Nielsen Norman Group carousel guidance.
17. **Expedia (reported by Bryan Eisenberg / UX literature)** — removing one redundant form field attributed with ~$12M annual profit impact.
18. **Sucuri (GoDaddy) Hacked Website Reports** — outdated/vulnerable plugins, themes and cores as the leading WordPress compromise vector; RevSlider/SoakSoak campaign (2014–15) compromised 100,000+ sites.

STR-WEB-2026-01 · Rev A · Audit method: live HTTP inspection of strucon.net + production build analysis of the replacement platform. Measured facts verifiable via page source; research figures per cited studies.